Calcurion Physics Worksheets

A-Level Physics

Topic: Nuclear Physics

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Define binding energy. **[2]**

2. What is the binding energy per nucleon a measure of? **[1]**

3. Sketch a binding energy per nucleon curve, labelling fission and fusion regions. **[3]**

4. Define the atomic mass unit (u). **[1]**

5. Convert 1 u to MeV. ($1\ \text{u} = 1.66 \times 10^{-27}$ kg, $c = 3.0 \times 10^{8}$) **[3]**

6. State the radioactive decay equation: $N = N_0 e^{-\lambda t}$. **[1]**

7. Define the decay constant λ. **[1]**

8. A sample has decay constant $0.010\ \text{s}^{-1}$. Calculate its half-life. **[2]**

9. Activity $A = \lambda N$. A sample has $N = 5.0 \times 10^{15}$ and $\lambda = 2.0 \times 10^{-6}\ \text{s}^{-1}$. Calculate activity. **[2]**

10. What is nuclear fission? **[2]**

11. What is nuclear fusion? **[2]**

12. State one advantage and one disadvantage of nuclear fission power. **[2]**

13. Write the equation for the alpha decay of $^{238}_{92}\mathrm{U}$. **[2]**

14. Write the equation for the beta-minus decay of $^{14}_{6}\mathrm{C}$. **[2]**

15. What is meant by background radiation? **[1]**

Section B – Hard (Q16–25)

16. Calculate the binding energy of $^{4}_{2}\mathrm{He}$ given mass defect 0.0304 u. (1 u = 931 MeV) **[2]**

17. Calculate the binding energy per nucleon for $^{4}_{2}\mathrm{He}$. **[2]**

18. A radioactive sample has half-life 8.0 days. What fraction remains after 24 days? **[2]**

19. Activity falls from 800 Bq to 100 Bq in 12 hours. Calculate the half-life. **[3]**

20. Explain how a chain reaction is controlled in a nuclear reactor. **[3]**

21. Calculate the energy released when 1.0 g of $^{235}\mathrm{U}$ undergoes fission, if 200 MeV is released per fission. ($N_A = 6.0 \times 10^{23}$) **[4]**

22. State the conditions required for nuclear fusion to occur. **[2]**

23. Explain why fusion is difficult to achieve on Earth. **[3]**

24. A nucleus has radius $R = R_0 A^{1/3}$. Calculate the radius of $^{12}\mathrm{C}$ if $R_0 = 1.2$ fm. **[2]**

25. Describe the process of induced fission of $^{235}\mathrm{U}$. **[3]**

End of Worksheet • Show all working clearly.